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Bolt feeder

Abstract

A bolt feeder includes: a rotating plate (12) arranged adjacent to a hopper (27) storing bolts (10); one or more magnets (15) to attract the bolts (10), from a back surface of the rotating plate (12), onto a front surface of the rotating plate (12); a selection block (13) having a gap space (13a) through which one bolt in a predetermined orientation out of the bolts (10) attracted by the one or more magnets (15) is allowed to pass; a motor (26) to cause the one or more magnets (15) to rotate and move from the hopper (27) toward the selection block (13); and a guide plate (29) extending upstream in a moving direction of the one or more magnets (15) from an entrance of the gap space (13a), and guiding the bolts (10) attracted by the one or more magnets (15) to the gap space (13a).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a U.S. National Stage Application under 35 U.S.C § 371 of International Patent Application No. PCT/JP2021/013263 filed Mar. 29, 2021, which claims the benefit of priority to Japanese Patent Application No. 2020-084230 filed May 13, 2020, the disclosures of all of which are hereby incorporated by reference in their entities.

TECHNICAL FIELD

(2) The present invention relates to a bolt feeder which aligns and feeds bolts in a predetermined orientation, to be used in a welding station or the like.

BACKGROUND ART

(3) When bolts are used in a welding station, for example, the bolts have to be aligned one by one in a predetermined orientation so as to be fed to the welding station. A bolt feeder required for this processing includes a selection block **13A** in a rectangular parallelepiped shape, as shown in FIG. **12**. The selection block **13A** has a gap space **13aA** wide enough for threaded portions **10b** of bolts **10** to pass through one by one. The gap space **13aA** is formed through the selection block **13A** from a front surface to a rear surface thereof in a feed direction of the bolts **10**.

(4) The selection block **13A** faces a rotating plate **12A** in a disk shape at a lower portion thereof, via a gap. The gap has a dimension slightly larger than a thickness of a head **10a** of the bolt **10**. When the threaded portion **10b** of the bolt **10** enters the gap space **13aA** upright with respect to the rotating plate **12A**, the head **10a** can enter the gap.

(5) The rotating plate **12A** rotates in a direction indicated by an arrow toward an entrance of the gap space **13aA**, and one or more magnets (not shown) are arranged on a back surface of the rotating plate **12A**. With the one or more magnets, the bolts **10** stored in a hopper (not shown) are attracted in small numbers. When the rotating plate **12A** rotates, the bolts **10** in the hopper are attracted by the one or more magnets and conveyed to the gap space **13aA**. Only one bolt, with the head **10a** facing the rotating plate **12A** and the threaded portion **10b** in an upright orientation, out of the conveyed bolts **10** passes through the selection block **13A**. Specifically, the bolt **10** is moved, with the threaded portion **10b** inserted in the gap space **13aA**, in an upright orientation, and the head **10a** inserted in the gap. The bolts **10** attracted in other orientations are blocked by the selection block **13A** and fall into the hopper (not shown). In the same manner, the other bolts **10** are sequentially inserted and moved one by one.

(6) Thus, the bolts **10** are sequentially inserted into the selection block **13A** one by one in the predetermined orientation and moved, and are then fed to the welding station (not shown) through a shoot rail **14A**. Patent Document 1 discloses such a conventional technique as described above.

CITATION LIST

Patent Document 1

(7) Patent Document 1: Japanese Patent NO. 3117662

SUMMARY OF THE INVENTION

Problems to be Solved

(8) However, the bolts **10** are attracted by the one or more magnets on the rotating plate **12A** and conveyed so that the bolts **10** may be densely attracted and conveyed. In this case, when the bolts **10** are conveyed to the gap space **13aA** of the selection block **13A** in a concentrated manner, the bolts **10** may be tangled with each other and stuck in the entrance of the gap space **13aA**, as shown in FIG. **12**. As one example of sticking, the threaded portion **10b** of the bolt **10** in a laid-down orientation is inserted into the gap space of the selection block **13A**, and another bolt **10** is caught between said bolt **10** and a wall surface **29a** of a guide plate **29**. In this case, an operator has to release the sticking of the bolts **10**, resulted in a problem such as a lower yield rate.

(9) The present invention is intended to solve the problem described above and to provide a bolt feeder which aligns bolts one by one in a predetermined orientation, without being stuck, and feeds the bolts to a subsequent station.

Solution to Problem

(10) To solve the problem described above, a bolt feeder according to the present invention is one to align bolts in a predetermined orientation and feed the bolts to a subsequent station, including: a storage portion storing the bolts therein; a plate-shaped body arranged adjacent to the storage portion; one or more magnets placed on a back surface of the plate-shaped body to attract the bolts via the plate-shaped body; a selection portion arranged on a front surface of the plate-shaped body and having a passage through which one bolt in a predetermined orientation out of the bolts attracted by the one or more magnets is allowed to pass; a moving device causing the one or more magnets, or the one or more magnets and the plate-shaped body, to rotate and move from the storage portion toward the selection portion; and a guide portion arranged on the front surface of the plate-shaped body, extending upstream in a moving direction of the one or more magnets from an entrance of the passage, and guiding the bolts attracted by the one or more magnets to the passage, wherein the guide portion has an inclined surface around the entrance of the passage, and the inclined surface is formed such that the more a position on the inclined surface is located away from a rotation center of the one or more magnets, the more the position is away from the front surface of the plate-shaped body.

Advantageous Effects of the Invention

(11) According to the present invention, bolts can be aligned one by one in a predetermined orientation, without being stuck, and fed to a subsequent station.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of a bolt feeder according to an embodiment of the present invention;
- (2) FIG. 2 is a partial cross-sectional side view of the bolt feeder;
- (3) FIG. 3 is a perspective view of the bolt feeder to show an inclined surface of a guide plate and a surrounding structure thereof;
- (4) FIG. 4 is a side view of a selection block, across a gap, and an upright bolt on a rotating plate;
- (5) FIG. 5 is a side view of the selection block, across the gap, and a bolt on the rotating plate, with a threaded portion laid down;
- (6) FIG. 6 is a side view of the selection block, across the gap, and a bolt on the rotating plate, with the threaded portion being raised;
- (7) FIG. 7 is a cross-sectional view, taken along a line A-A in FIG. 3;
- (8) FIG. 8 is a perspective view of bolts attracted by one or more magnets on the rotating plate;
- (9) FIG. 9 is a perspective view of a bolt, with a threaded portion entered a gap space in the selection block, tangled with another bolt;
- (10) FIG. 10 is a perspective view of a bolt, with an upright threaded portion entered the gap space of the selection block;
- (11) FIG. 11 is a perspective view of bolts, with threaded portions entered the gap space of the selection block and stuck; and,
- (12) FIG. 12 is a perspective view of main parts of a conventional bolt feeder.

DETAILED DESCRIPTION OF EMBODIMENTS

Configuration of Embodiment

- (13) Hereinbelow, a description is given of an embodiment of the present invention with reference to drawings. FIG. 1 is a front view of a bolt feeder, and FIG. 2 is a side view of the bolt feeder, according to an embodiment of the present invention.
- (14) A bolt feeder 20 in FIG. 1 includes a control board 22 having a control panel 21 and arranged vertically on a floor, and a bolt feeding part 23 arranged on the control board 22.
- (15) As shown in FIG. 2, the bolt feeding part 23 includes a base 24 in an L-shape in a side view,

having an inclined plate **24a** formed with a circular opening **24b** (see FIG. 1) therein, a rotating plate **12** in a disk shape (plate-shaped body) rotatably assembled at the circular opening **24b**, and a motor (moving device) **26** with a rotating shaft **26a** fixed in the center of the rotating plate **12**. The motor **26** is covered by a motor cover **28**.

(16) Further, the bolt feeding part **23** includes a hopper (storage portion) **27** arranged on a part of the inclined plate **24a**, closer to the control panel **21**, so as to protrude upward at an angle with respect to the inclined plate **24a**, a shoot rail **14** protruding laterally downward from the base **24** and further extending downward, as shown in FIG. 1, and a proximity sensor **31** attached to the shoot rail **14**. Furthermore, the bolt feeding part **23** includes, as shown in FIG. 3, a selection block (selection portion) **13** in a rectangular parallelepiped shape connected to the uppermost portion of the shoot rail **14**, and a guide plate (guide portion) **29** fixed to the inclined plate **24a**. Note that the proximity sensor **31** is a sensor in the appended one or more claims.

(17) The selection block **13** has a gap space (passage) **13a** through which the bolts **10** pass. The selection block **13** has the gap space **13a** formed therethrough from a front surface (or surface facing the bolts) thereof to a rear surface thereof in a feeding direction of the bolts **10**. Further, the gap space **13a** has a dimension wide enough for one or more threaded portions **10b** of the bolts **10** passing through one by one, or one after another, as shown in FIG. 4. The rotating plate **12** is arranged below the gap space **13a** and the selection block **13**, across a gap (interval) **G1**.

(18) The gap **G1** has a dimension slightly larger than a thickness **t1** of a head **10a** of the bolt **10**. That is, when the threaded portion **10b** of the single bolt **10** enters the gap space **13a**, in an upright orientation allowed to pass through, the head **10a** enters the gap **G**, in an orientation allowed to pass through.

(19) As indicated by an arrow **Y1** in FIG. 3, the rotating plate **12** rotates toward an entrance of the gap space **13a**, and magnets **15** are placed on a periphery of a back surface of the rotating plate **12** at angular intervals of 90 degrees. However, one or more magnets **15** may be placed at one or more positions on the back surface to pass across the gap space **13a** via the rotating plate **12** and the gap **G**, when rotated. Some of a large number of bolts **10** stored in the hopper **27** are attracted by the magnets **15** on a front surface of the rotating plate **12** and conveyed to the gap space **13a**.

(20) As shown in FIG. 4, lower corners **13b** on a front side of the selection block **13** each have a rounded shape in a side view. The lower corners **13b** in a rounded shape (rounded lower corners) serve to raise the threaded portion **10b** of the laid-down bolt **10** so as to stand upright. For example, as shown in FIG. 5, when the threaded portion **10b** of the laid-down bolt **10** attracted by the magnet **15** on the rotating plate **12** enters the gap space **13a** and the gap **G1**, the head **10a** is in contact with the front surface of the selection block **13** and the rotating plate **12** at an angle.

(21) In this state, when the rotating plate **12** further rotates in the direction of the arrow **Y1**, the head **10a** of the bolt **10** moves in the direction of the arrow **Y1** while being in contact with the rounded lower corners **13b**, as shown in FIG. 6. In this movement, a leading end in a moving direction of the head **10a** moves into the gap **G1** and a trailing end of the head **10a** moves to come closer to the rotating plate **12**. Therefore, the threaded portion **10b** protruding from the head **10a** moves toward an upright orientation as indicated by an arrow **Y2**.

(22) As the bolts **10** further moves, the threaded portion **10b** of the bolt **10** stands upright, and the head **10a** is in an orientation to enter the gap **G1**, as shown in FIG. 4. In this way, the bolts **10** are inserted and moved into the space **13a** of the selection block **13** one by one in a predetermined orientation and are fed to a subsequent station through the shoot rail **14** in FIG. 3. The shoot rail **14** is connected, at the uppermost portion thereof, to the rear surface of the gap space **13a** of the selection block **13**, protrudes laterally downward of the base **24** from the connected portion, and further extends downward, as shown in FIG. 1. Therefore, the bolts **10** having passed through the gap space **13a** freely fall through the shoot rail **14**.

(23) Next, the guide plate **29** shown in FIG. 3 is arranged along the outer periphery of the rotating plate **12** on the entrance side (front side) of the gap space **13a** of the selection block **13**. The guide

plate **29** serves to guide the bolts **10** (see FIG. **8**), which are attracted by the one or more magnets **15** on the rotating plate **12**, to the gap space **13a**. For the guiding, the guide plate **29** has a curved end-surface **29a** which is a wall surface having a predetermined height and extending in a curve along the outer periphery of the rotating plate **12**. The curved end-surface **29a** is the wall surface perpendicular to or nearly perpendicular to the plane of the rotating plate **12**.

(24) FIG. **7** is a cross-sectional view, taken along a line A-A in FIG. **3**. As shown in FIG. **7**, the curved end-surface **29a** has an inclined surface **29b** which inclines upward at a predetermined angle θ_1 from the plane of the rotating plate **12** toward a direction perpendicular to (intersecting) the curved end-surface **29a**, around the entrance of the gap space **13a**. In other words, the inclined surface **29b** is inclined such that the more a position on the inclined surface **29b** is located away from a rotation center of the one or more magnets **15** (outward in the radial direction), the more the position is away from (in a direction normal to) the front surface of the rotating plate **12**. The angle θ_1 is set at an angle of less than 90 degrees with respect to the back surface of the guide plate **29**, which is flush with the plane of the rotating plate **12**. In the present example, the angle θ_1 is set to 45 degrees.

(25) The inclined surface **29b** has, as shown in FIG. **3**, a rounded surface **9b1** formed in a rounded shape at a portion thereof closer to the entrance of the gap space **13a**, and a similar rounded surface **9b2** on the opposite side. That is, the inclined surface **29b** has the rounded surfaces **9b1** and **9b2** at both ends thereof, which are continuous to the inclined surface. In this structure, the rounded surface **9b1** is formed at the entrance of the gap space **13a**, to facilitate the bolts **10** changing orientations and escaping by the rounded surface **9b1**. Therefore, even when the bolts **10** are conveyed to the entrance of the gap space **13a** in a concentrated manner, the bolts **10** other than that inserted into the gap space **13a** can have orientations thereof changed and escaped by the rounded surface **9b1**.

(26) Note that, the inclined surface **29b** may have only the rounded surface **9b1**, closer to the entrance of the gap space **13a**, of the rounded surfaces **9b1** and **9b2** at both ends. Further, the rounded surfaces **9b1** and **9b2** at both ends may each have a right-angled shape or any other polyangular shape.

(27) Thus, when the curved end-surface **29a** has the inclined surface **29b**, an angle θ_2 between the inclined surface **29b** and the plane of the rotating plate **12** (also called a rotation plane) is an obtuse angle larger than an angle θ_3 (90 degrees) between the vertical curved end-surface **29a** and the rotation plane, as shown in FIG. **7**. In the present embodiment, the obtuse angle θ_2 is 135 degrees, which is the sum of 90 degrees of the angle θ_3 and 45 degrees of the angle θ_1 .

(28) This defines an obtuse-angled opening open at an angle of 135 degrees around the entrance of the gap space **13a**, to facilitate the bolts **10** gathered around the entrance easily being ejected away from the entrance through the obtuse-angled opening. For example, when an inverted bolt **10B** interposed between a laid-down bolt **10A** and the curved end-surface **29a** is conveyed, as shown in FIG. **8**, the threaded portion **10b** of the bolt **10A** is inserted into the gap space **13a** and the other bolt **10B** is moved upward along the inclined surface **29b**, as shown in FIG. **9**. Accordingly, the other bolt **10B** is ejected away from the entrance of the gap space **13a**, without being held between the bolt **10A** and the inclined surface **29b**.

(29) Further, as shown in FIG. **3**, the rounded surface **9b1** is formed in the inclined surface **29b** at the entrance of the gap space **13a**, so that even when the bolts **10** are conveyed in a concentrated manner at the entrance of the gap space **13a**, most of the bolts **10** are changed in orientation by the rounded surface **9b1** and are easily ejected.

(30) Next, as shown in FIG. **11**, a description is given of a structure to rotate the rotating plate **12** reversely in a direction of an arrow **Y3** when the bolts **10** are stuck in the gap space **13a**.

(31) The structure includes the proximity sensor **31** in FIG. **1**, the motor **26** in FIG. **2** which causes the rotating plate **12** to rotate forward (in the direction of the arrow **Y1** in FIG. **3**) or reversely, and a controller **33** with a timer **32**. The proximity sensor **31**, the timer **32**, the controller **33**, and the

motor **26** are electrically connected with each other.

(32) The proximity sensor **31** detects the bolt **10** made of metal passing through the shoot rail **14** and outputs a detection signal to the timer **32**. The timer resets timing operation when the detection signal is inputted to perform the timing operation again. In a case where no detection signal is inputted to the timer **32**, the timing operation continues. When the timing operation has elapsed a predetermined first setting time, the timer **32** outputs a no-detection signal, indicating that the bolt **10** has not been detected, to the controller **33**.

(33) In other words, the timer **32** measures time of the bolt **10** not being detected by the proximity sensor **31** and outputs the no-detection signal to the controller **33** when the time of the bolts **10** not being detected has elapsed the first setting time.

(34) When start-up operation is performed with the control panel **21**, the controller **33** controls the motor **26** to rotate forward. During this controlling, when the no-detection signal is inputted from the timer **32**, the controller **33** controls the motor **26** to rotate reversely. In other words, when the timing operation of the timer **32** has elapsed the first setting time, the controller **33** controls the motor **26** to rotate reversely. With this control, the motor **26** rotates reversely to cause the rotating plate **12** to rotate reversely (in the direction of the arrow **Y3**), so that the bolts **10** stuck in the gap space **13a** shown in FIG. **11** are released.

(35) Further, when a predetermined time (second setting time) elapses since the motor **26** has been controlled to rotate reversely, the controller **33** controls the motor **26** to rotate forward (in the opposite direction of the arrow **Y3**). The second setting time defines, by measurement or the like in advance, time for the bolts **10** stuck in the gap space **13a** to be released by reverse rotation of the rotating plate **12**. The second setting time is determined by the controller **33** with a built-in timer or measured by the timer **32** separately from the first setting time. Note that the second setting time is a setting time in the appended one or more claims.

Operation of Embodiment

(36) Next, a description is given of feeding operation of the bolts **10** by the bolt feeder **20** described above. Note that a large number of bolts **10** is stored in the hopper **27**.

(37) First, when an operator performs the start-up operation with the control panel **21**, the controller **33** controls the motor **26** to rotate forward, which, in turn, causes the rotating plate **12** to rotate forward by the motor **26** rotating forward. When the rotating plate **12** rotates forward, the bolts **10** attracted by the magnets **15** onto the front surface of the rotating plate **12** out of the bolts **10** in the hopper **27** are conveyed, as shown in FIG. **8**, toward the gap space **13a** of the selection block **13**.

(38) Then, when the bolts **10** are conveyed, as shown in FIG. **9**, simultaneously to the entrance of the gap space **13a** in a concentrated manner, the threaded portion **10b** of one of the bolts **10** may enter the gap space **13a** and may be tangled with the threaded portion **10b** of the other bolt **10** at the entrance of the gap space. The head **10a** of the other bolt **10** is located on the rounded surface **9b1** of the inclined surface **29b**. Therefore, when the rotating plate **12** further rotates forward, the other bolt **10** is moved away laterally with respect to the gap space **13a** from the obtuse-angled opening defined by the rounded surface **9b1**.

(39) After the other bolt is moved away, the bolt **10** with the threaded portion **10b** in the gap space **13a** is in an orientation shown in FIG. **5**, by movement of forward rotation of the rotating plate **12**. That is, the head **10a** stands at an angle between the front surface of the selection block **13** and the rotating plate **12**, with the laid-down threaded portion **10b** in the gap space **13a** and the gap **G1**.

(40) Then, when the rotating plate **12** further rotates forward, as shown in FIG. **6**, the head **10a** of the bolt **10** moves in the direction indicated by the arrow **Y1** while being in contact with the rounded lower corners **13b**, to cause the threaded portion **10b** protruding from the head **10a** to move toward the direction for upright orientation indicated by the arrow **Y2**.

(41) When the bolt **10** further moves, as shown in FIG. **10**, the threaded portion **10b** of the bolt **10** stands upright and the head **10a** is in an orientation to enter the gap **G1**. The bolt **10** in this

orientation passes through the gap space **13a**, moves along the shoot rail **14** in free fall, and is fed to a subsequent station. Then, the bolts **10** sequentially pass through the gap space **13a** one by one in a predetermined orientation and are fed to the subsequent station through the shoot rail **14** in a same manner.

(42) In contrast, even when the rotating plate **12** rotates forward, the bolts **10** may be stuck in the gap space **13a**, as shown in FIG. **11**, one of the bolts **10** cannot pass through the gap space **13a** and cannot move along the shoot rail **14**. Therefore, the proximity sensor **31** does not detect the bolt **10** and a detection signal is not outputted to the timer **32**.

(43) At this time, the timer **32** is not inputted with the detection signal and continues timing operation. When the timing operation has elapsed the first setting time, the timer **32** outputs a no-detection signal to the controller **33**. When the no-detection signal is inputted, the controller **33** controls the motor **26** to rotate reversely. With this control, the motor **26** rotates reversely to cause the rotating plate **12** to rotate reversely, to move away the bolts **10** stuck in the gap space **13a** shown in FIG. **11**.

(44) Then, when the second setting time elapses since the above-mentioned reverse rotation control has been performed, the controller **33** controls the motor **26** to rotate forward. With this control, the motor **26** rotates forward and the rotating plate **12** rotates forward, so that the bolts **10** attracted by the one or more magnets **15** on the rotating plate **12** sequentially pass through the gap space **13a** one by one in a predetermined orientation and are fed through the shoot rail **14** to the subsequent station.

Advantageous Effects of Embodiment

(45) As described above, the bolt feeder **20** of the present embodiment aligns and feeds the bolts **10** one by one in a predetermined orientation to a subsequent station, to have following features.

(46) (i) The bolt feeder **20** includes: the hopper **27** storing the bolts **10** therein; the rotating plate **12** arranged adjacent to the hopper **27**; the one or more magnets **15** placed on the back surface of the rotating plate **12** to attract the bolts **10** via the rotating plate **12**; and the selection block **13** arranged on the front surface of the rotating plate **12** and having the gap space **13a** through which the bolt **10** in a predetermined orientation out of the bolts **10** attracted by the one or more magnets **15** is allowed to pass.

(47) Further, the bolt feeder **20** includes: the motor **26** causing the one or more magnets **15** and the rotating plate **12** to rotate and move from the hopper **27** toward the selection block **13**; and the guide plate **29** arranged on the front surface of the rotating plate **12**, extending upstream in the moving direction of the one or more magnets **15** from the entrance of the gap space **13a**, and guiding the bolts **10** attracted by the one or more magnets **15** to the gap space **13a**.

(48) The guide plate **29** has the inclined surface **29b** around the entrance of the gap space **13a**, and the inclined surface **29b** is formed such that the more a position on the inclined surface **29b** is located away from the rotation center of the one or more magnets **15** (outward in the radial direction), the more the position is away from the front surface of the rotating plate **12** (in the direction normal to the front surface).

(49) With the structure, the following advantageous effects are obtained. The curved end-surface **29a** has the inclined surface **29b**, which inclines upward from the plane of the rotating plate **12** toward a direction intersecting the curved end-surface **29a**, around the entrance of the gap space **13a**. The upward inclination of the inclined surface **29b** is set to the angle $\theta 1$ (45 degrees) as shown in FIG. **7** with respect to the back surface of the inclined surface **29b** which is flush with the plane of the rotating plate **12**, for example. Also, the angle $\theta 2$ between the inclined surface **29b** and the plane of the rotating plate **12** (rotation plane) is set to the obtuse angle (90 degrees+45 degrees=135 degrees) larger than the angle $\theta 3$ (90 degrees) between the curved end-surface **29a** and the rotation plane. Therefore, the obtuse-angled opening having the angle $\theta 2$ is defined around the entrance of the gap space **13a**.

(50) In a conventional art, an opening corresponding to the obtuse-angled opening is set to an angle

of only 90 degrees between the rotation plane and the curved end-surface **29a**, and both the curved end-surface **29a**, defining the opening at the angle of 90 degrees, and vertical wall surfaces in the opening of the gap space **13a** are close to each other and upright with respect to the rotating plate **12**, so that the bolts **10** are easily caught and stuck between the two.

(51) In contrast, in the present invention, the obtuse-angled opening at the angle of 135 degrees, which exceeds the angle $\theta 3$ of 90 degrees, is widely opened between the rotation plane and the inclined surface **29b**. That is, a distance between the vertical wall surfaces in the opening of the gap space **13a** and the inclined surface **29b** is increased. This allows the bolts **10**, conveyed to the entrance of the gap space **13a** in a concentrated manner, to be moved away from the obtuse-angled opening outward around the entrance. Therefore, the bolts **10** are not stuck so that the bolts **10** conveyed to the gap space **13a** are aligned one by one in a predetermined orientation, without being stuck, and fed to a subsequent station.

(52) (ii) The bolt feeder **20** includes: the hopper **27** storing the bolts **10** therein; the rotating plate **12** arranged adjacent to the hopper **27**; the one or more magnets **15** arranged on the back surface of the rotating plate **12** to attract the bolts **10** via the rotating plate **12**; and the selection block **13** arranged on the front surface of the rotating plate **12** and having the gap space **13a** through which the bolt **10** in a predetermined orientation out of the bolts **10** attracted by the one or more magnets **15** is allowed to pass.

(53) Further, the bolt feeder **20** includes: the motor **26** causing the one or more magnets **15** and the rotating plate **12** to rotate and move from the hopper **27** toward the selection block **13**; and the guide plate **29** arranged on the front surface of the rotating plate **12**, extending upstream in the moving direction of the one or more magnets **15** from the entrance of the gap space **13a**, and guiding the bolts **10** attracted by the one or more magnets **15** to the gap space **13a**.

(54) Still further, the bolt feeder **20** includes: the proximity sensor **31** which detects the bolt **10** having passed through the selection block **13**; and the controller **33** which controls the motor **26** as a moving device to rotate reversely when the proximity sensor **31** does not detect the bolt **10** for a predetermined time.

(55) With the structure, as shown in FIG. **11**, in a case where the rotating plate **12** rotates forward (in the opposite direction of the arrow **Y3**) but the bolts **10** are stuck in the gap space **13a** and cannot pass through the gap space **13a**, the bolts **10** do not pass through the shoot rail **14**. Therefore, if no bolt **10** is detected by the proximity sensor **31** for a predetermined time, the controller **33** controls the motor **26** to rotate reversely. With this control, the motor **26** rotates reversely to cause the rotating plate **12** to rotate reversely (in the direction of the arrow **Y3**), so that the bolts **10** stuck in the gap space **13a** are released.

(56) (iii) The bolt feeder **20** includes: the hopper **27** storing the bolts **10** therein; the rotating plate **12** arranged adjacent to the hopper **27**; the one or more magnets **15** arranged on the back surface of the rotating plate **12** to attract the bolts **10** via the rotating plate **12**; and the selection block **13** arranged on the front surface of the rotating plate **12** and having the gap space **13a** through which the bolt **10** in a predetermined orientation out of the bolts **10** attracted by the one or more magnets **15** is allowed to pass.

(57) Further, the bolt feeder **20** includes: the motor **26** causing the one or more magnets **15** and the rotating plate **12** to rotate and move from the hopper **27** toward the selection block **13**; and the guide plate **29** arranged on the front surface of the rotating plate **12**, extending upstream in the moving direction of the one or more magnets **15** from the entrance of the gap space **13a**, and guiding the bolts **10** attracted by the one or more magnets **15** to the gap space **13a**.

(58) The guide plate **29** described above has the inclined surface **29b** around the entrance of the gap space **13a**, and the inclined surface **29b** is formed such that the more a position on the inclined surface **29b** is located away from the rotation center of the one or more magnets **15** (outward in the radial direction), the more the position is away from the front surface of the rotating plate **12** (in the direction normal to the front surface).

(59) Still further, the bolt feeder **20** includes: the proximity sensor **31** which detects the bolt **10** having passed through the selection block **13**; and the controller **33** which controls the motor **26** as a moving device to rotate reversely when the proximity sensor **31** does not detect the bolt **10** for a predetermined time.

(60) With the structure, the same advantageous effects as (i) and (ii) described above are obtained.

(61) (iv) When the setting time elapses since the controller **33** has controlled the motor **26** to rotate reversely, the controller **33** controls the motor **26** to rotate forward.

(62) With the structure, the following advantageous effects are obtained. The setting time is set in advance to a time taken to release the bolts **10** stuck in the gap space **13a** by reverse rotation of the rotating plate **12**. When the motor **26** rotates reversely for the setting time, the bolts **10** stuck in the gap space **13a** are released. When the setting time has elapsed, the controller **33** controls the motor **26** to rotate forward, to return to normal operation. The bolts **10** attracted by the one or more magnets **15** on the rotating plate **12** sequentially pass through the gap space **13a** one by one in a predetermined orientation, and are fed through the shoot rail **14** to a subsequent station.

(63) (v) The selection block **13** has a pair of walls arranged across a gap space (distance) **13a** corresponding to the threaded portion **10b** of the bolt **10**, and the walls are arranged across the gap **G1** corresponding to the head **10a** of the bolt **10** with respect to the rotating plate **12**. The wall has, at the end thereof closer to the rotating plate **12**, a rounded shape (curved shape) in cross section when viewed radially from the rotation center of the one or more magnets **15**.

(64) With the structure, the following advantageous effects are obtained. As shown in FIG. 5, the laid-down threaded portion **10b** of the bolt **10** may enter the gap space **13a** of the selection block **13** and the gap **G1**, and the head **10a** may stand at an angle between the front surface of the selection block **13** and the rotating plate **12**.

(65) With the bolt **10** in this orientation, when the rotating plate **12** further rotates in the direction indicated by the arrow **Y1**, as shown in FIG. 6, the head **10a** moves in the gap **G1** while being in contact with the rounded lower corners **13b**, and the threaded portion **10b** protruding from the head **10a** stands up in the direction of the arrow **Y2**. When the rotating plate **12** rotates further, as shown in FIG. 4, the threaded portion **10b** of the bolt **10** stands upright, and the head **10a** is in an orientation to enter the gap **G1**. That is, even when the bolt **10**, with the threaded portion **10b** in a laid-down orientation, enters the gap space **13a**, the bolt **10** in this orientation can be arranged in a predetermined orientation (the head **10a** is down and the threaded portion **10** is upright) and pass through.

(66) The present invention may be appropriately modified into other specific configurations within a scope of the present invention. Instead of the rotating plate **12**, a plate-shaped body may be fixed and only the one or more magnets **15** may be rotated. Further, instead of the curved end-surface **29a** of the guide plate **29**, a non-curved wall surface may be used. The non-curved wall surface may be, for example, a straight wall surface inclined upward toward the selection block **13**. In addition, instead of the shoot rail **14**, a holder may be used to hold the aligned bolts **10**.

REFERENCE SYMBOLS

(67) **9b1**, **9b2**: rounded surface; **10**: bolt; **10a**: threaded portion; **10b**: head; **12**: rotating plate; **13**: selection block; **13a**: gap space; **13b**: rounded lower corner; **14**: shoot rail; **15**: magnet; **20**: bolt feeder; **26**: motor; **29**: guide plate; **29a**: curved end-surface; **29b**: inclined surface; **31**: proximity sensor; **32**: timer; **33**: controller.

Claims

1. A bolt feeder to align bolts in a predetermined orientation and to feed the bolts to a subsequent station, comprising: a storage portion storing the bolts therein; a plate-shaped body arranged adjacent to the storage portion; one or more magnets placed on a back surface of the plate-shaped body to attract the bolts via the plate-shaped body; a selection portion arranged on a front surface

of the plate-shaped body and having a passage through which one bolt in a predetermined orientation out of the bolts attracted by the one or more magnets is allowed to pass; a moving device causing the one or more magnets, or the one or more magnets and the plate-shaped body, to rotate and move from the storage portion toward the selection portion; and a guide portion arranged on the front surface of the plate-shaped body, extending upstream in a moving direction of the one or more magnets from an entrance of the passage, and guiding the bolts attracted by the one or more magnets to the passage, wherein the guide portion has an inclined surface around the entrance of the passage, and the inclined surface is inclined upward and radially outward from an end of the front surface of the plate-shaped body.

2. The bolt feeder according to claim 1, wherein the selection portion has a pair of walls arranged across a gap space corresponding to a threaded portion of the bolt, the walls are arranged across a gap corresponding to a head of the bolt with respect to the plate-shaped body, and the walls each have, at an end thereof closer to the plate-shaped body, a curved shape in cross section when viewed radially from a rotation center of the one or more magnets.

3. A bolt feeder to align bolts in a predetermined orientation and to feed the bolts to a subsequent station, comprising: a storage portion storing the bolts therein; a plate-shaped body arranged adjacent to the storage portion; one or more magnets placed on a back surface of the plate-shaped body to attract the bolts via the plate-shaped body; a selection portion arranged on a front surface of the plate-shaped body and having a passage through which one bolt in a predetermined orientation out of the bolts attracted by the one or more magnets is allowed to pass; a moving device causing the one or more magnets, or the one or more magnets and the plate-shaped body, to rotate and move from the storage portion toward the selection portion; and a guide portion arranged on the front surface of the plate-shaped body, extending upstream in a moving direction of the one or more magnet from an entrance of the passage, and guiding the bolts attracted by the one or more magnets to the passage, wherein the guide portion has an inclined surface around the entrance of the passage, the inclined surface is inclined upward and radially outward from an end of the front surface of the plate-shaped body, and the bolt feeder further comprises: a sensor to detect the bolt having passed through the selection portion; and a controller to control the moving device to rotate reversely when the sensor does not detect the bolt in a predetermined time.

4. The bolt feeder according to claim 3, wherein, when a setting time set in advance elapses since the controller has controlled the moving device to rotate reversely, the controller controls the moving device to rotate forward.

5. The bolt feeder according to claim 4, wherein the selection portion has a pair of walls arranged across a gap space corresponding to a threaded portion of the bolt, the walls are arranged across a gap corresponding to a head of the bolt with respect to the plate-shaped body, and the walls each have, at an end thereof closer to the plate-shaped body, a curved shape in cross section when viewed radially from a rotation center of the one or more magnets.

6. The bolt feeder according to claim 3, wherein the selection portion has a pair of walls arranged across a gap space corresponding to a threaded portion of the bolt, the walls are arranged across a gap corresponding to a head of the bolt with respect to the plate-shaped body, and the walls each have, at an end thereof closer to the plate-shaped body, a curved shape in cross section when viewed radially from a rotation center of the one or more magnets.
